Optimization for Deep Learning (OPT4DL)

Abolfazl Hashemi, Purdue ECE

Second-Order Guarantees

Higher Quality Solutions



- Consider our training problem

$$\min_w f(w) := \mathbb{E}[\ell(w, z)] \tag{1}$$

- We showed SGD with update $w_{t+1} = w_t - \eta g_t$ satisfies $\mathbb{E}\|\nabla f(\hat{w})\|^2 \leq \epsilon$ using $T = \Omega(\epsilon^{-2})$ iterations
- This is called a first-order solution and in principle can be a local minimum, local maximum, or a saddle point
- Example: $f_1(w) = \frac{1}{3}w^3 - w$

Higher-order Solutions (Cont'd)



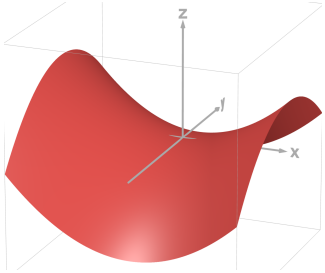
- Example: $f_2(w) = \frac{1}{3}w^3$

- Example: $f_3(w) = \frac{1}{4}w^4$

- Example: $f_4(w) = \frac{1}{2}(w^2 - 1)^2$

2nd Order Solutions

- Example: $f(x, y) = x^2 - y^2$



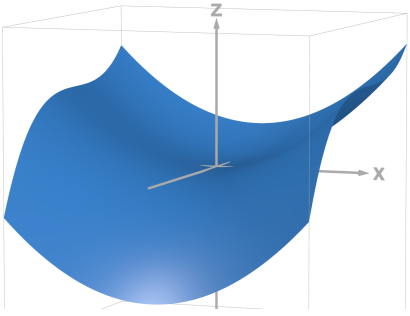
Definition

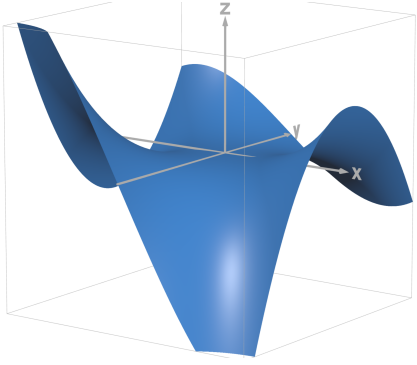
An (ϵ_1, ϵ_2)-accurate second order stationary solution for f is any $\hat{w}$ such that

$$\|\nabla f(\hat{w})\|^2 \leq \epsilon_1, \qquad \lambda_{\min}(\nabla^2 f(\hat{w})) \geq -\epsilon_2 \tag{2}$$

Which Solutions?

- The definition excludes local maximum and simple/strict saddle points (detectable by looking at Hessian).
- It does not however exclude higher order saddle points (undetectable by looking at Hessian)
- Examples: $x^2 + y^3$ (left) and monkey saddle, i.e. $x^3 - 3xy^2$ (right)



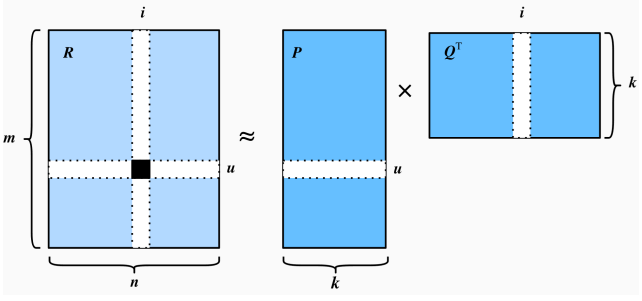


Why a Good Definition

- Many ML problems only include local/global minima and simple saddle points
- Example: Matrix Factorization

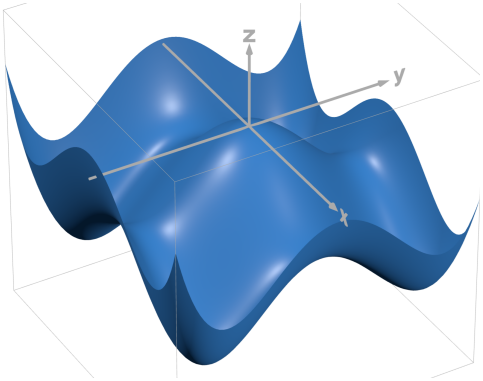
$$P^*, Q^* = \arg\min_{U,V} \|R - PQ^\top\|_F^2 \tag{3}$$

- Permutation of columns of P^* and Q^* leads to the same solution
- Similar idea applies to DL models and permutations weight matrices
- These exponentially many global minima are separated by exponentially many simple saddle points



Exponentially Many Saddles and Minima

- Consider the function $h(x) = (1 - x^2)^2$ and from it construct $h(x, y) = f(x) + f(y)$
  - 4 minima and 4 simple saddle points
  - $h(x, y, z) = f(x) + f(y) + f(z)$ will have 8 minima and 8 simple saddle points



Finding 2nd order Solutions



- Recall to find a first-order solution
  - We required accessing the gradient or a good approximation of that
  - We imposed smoothness, i.e., Lipschitzness of gradient
  - gradient is a first-order quantity
- Thus, intuitively, we expect to need the following
  - We require accessing the Hessian or a good approximation of that
  - We impose 2nd-order smoothness, i.e., smoothness of gradient, or Lipschitzness of Hessian
  - Hessian is a second-order quantity
- We will indeed require Lipschitzness of Hessian. However, we will discuss two methods one requiring Hessian and one requiring only (stochastic gradient)
- Hessian information will be useful to detect saddles and avoid them. Gradient noise injection could help avoid saddle points by taking advantage of their instability.
- In both cases, by avoid we mean find a direction to decrease the function value.

Second-Order Methods: Newton Algorithm



- Newton method is a celebrated algorithm for calculating zeros of functions iteratively:

$$f(x^*) = 0 \qquad x_{t+1} = x_t - \frac{f(x_t)}{f'(x_t)} \tag{4}$$

  Implication for optimization: finding zeros of gradient: $\nabla f(w^*) = 0$

$$w_{t+1} = w_t - H_t^{-1} \nabla f(w_t), \qquad H_t = \nabla^2 f(w_t) \tag{5}$$

- Newton Variations
  - Damped Newton: $H_t^{-1} \to \eta_t H_t^{-1}$
  - Levenberg–Marquardt Regularization (LMR) $H_t^{-1} \to (\eta_t^{-1} I_d + H_t)^{-1} \approx \eta_t (I_d - \eta_t H_t)$
- When $H_t^{-1} \to \eta I_d$ we obtain gradient descent.
- Effectively improves the local landscape conditions by leveraging the preconditioning matrix $C_t = H_t^{-1}$

Newton, GD, and Majorization Minimization



- Recall, we previously discussed GD as an example of MM for L-smooth functions when $\eta \leq 1/L$

$$w_{t+1} = \arg\min_{w} f(w_t) + \langle \nabla f(w_t), w - w_t \rangle + \frac{1}{2\eta}\|w - w_t\|^2 \tag{6}$$

- We can easily show Newton update is equivalent to minimizing the 2nd-order Taylor approximation at w_t

$$w_{t+1} = \arg\min_{w} f(w_t) + \langle \nabla f(w_t), w - w_t \rangle + \frac{1}{2}(w - w_t)^\top H_t (w - w_t) \tag{7}$$

- Note $(w - w_t)^\top H_t (w - w_t) \leq L\|w - w_t\|^2$ meaning Newton is not an instance of MM.
- Intuitively, Newton is designed to solve $\nabla f(w) = 0$ not $\nabla f(w) = 0$ such that $\nabla^2 f(w) \succeq 0$
- For this reason, Newton may not help us to avoid saddles effectively. But a variation of it will.
- Idea is to form an upper-bound and minimize that iteratively by assuming and leveraging Lipschitzness of Hessian as opposed to Lipschitzness of gradient.



Definition

A function f is ρ second-order smooth if $\|\nabla^2 f(w_1) - \nabla^2 f(w_2)\|_2 \leq \rho \|w_1 - w_2\|$ for all $w_1, w_t \in \mathbb{R}^d$

- It turns out this is equivalent to

$$f(w_1) \leq f(w_2) + \langle \nabla f(w_2), w_1 - w_2 \rangle + \frac{1}{2}(w_1 - w_2)^\top \nabla^2 f(w_2)(w_1 - w_2) + \frac{\rho}{6}\|w_1 - w_2\|^3 \quad (8)$$

$$\|\nabla f(w_1) - \nabla f(w_2) - \nabla^2 f(w_2)(w_1 - w_2)\| \leq \frac{\rho}{2}\|w_2 - w_1\|^2 \quad (9)$$

- Function has local cubic bound and the gradient is ρ-smooth (it has quadratic bound)
- Cubic Regularization of Newton Method (CRNM):

$$w_{t+1} = \arg\min_w f(w_t) + \langle \nabla f(w_t), w - w_t \rangle + \frac{1}{2}(w - w_t)^\top H_t (w - w_t) + \frac{\rho}{6}\|w - w_t\|^3 \quad (10)$$

- Interestingly, the update is equivalent to a convex optimization problem when $d \geq 2$

CRNM Idea

- Under ρ-Lipschitz Hessian Hessian, the method is MM thus, we continually reduce the function value
- Consider the spectral decomposition of Hessian $H_t = \sum_{i=1}^{d} \lambda_i u_i u_i^\top$
- So the update can be written equivalently as

$$w_{t+1} = \arg\min_{w} f(w_t) + \langle \nabla f(w_t), w - w_t \rangle + \frac{1}{2} \sum_{i=1}^{d} \lambda_i \left(u_i^\top (w - w_t) \right)^2 + \frac{\rho}{6} \|w - w_t\|^3 \quad (11)$$

  - We are penalizing directions u_i with positive λ_i as they are the direction of increase.
  - We are taking advantage of directions u_i with negative λ_i as they are the direction of decrease.

Theorem

CRNM can find an $(\epsilon, 2\sqrt{\rho\epsilon})$-second order solution in $T = \Omega(\frac{\sqrt{\rho}(f(w_1) - f^*)}{\epsilon^{0.75}})$.

Proof Sketch



- If there exists a direction with sufficiently large (and negative) eigen value such that $\lambda < -\sqrt{\rho\epsilon}$, we can reduce the function value

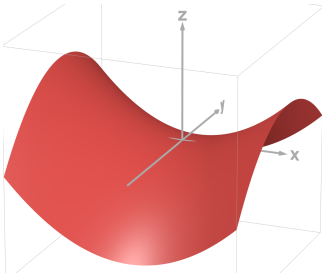
$$f(w_{t+1}) - f(w_t) \leq -\frac{2\lambda^3}{3\rho^2} \qquad \text{compare with} \qquad f(w_{t+1}) - f(w_t) \leq -\frac{2\|\nabla f(w_t)\|^2}{2L} \qquad \text{for GD} \tag{12}$$

- Otherwise, there are two cases. First, if $\|w_{t+1} - w_t\| \leq \epsilon\rho^{-1}$ then $\|\nabla f(w_{t+1})\|^2 \leq \epsilon$ meaning both conditions of approximate 2nd order definition are met and we find a solution.
- Secondly, if $\|w_{t+1} - w_t\| > \epsilon\rho^{-1}$, we can still make progress and reduce the function value

$$f(w_{t+1}) - f(w_t) < -\frac{\epsilon^{0.75}}{6\sqrt{\rho}} \tag{13}$$

- By contradiction we then show a solution is found after precisely $T = \Omega(\frac{\sqrt{\rho}(f(w_1) - f^*)}{\epsilon^{0.75}})$ iterations.

Avoiding Saddles with Perturbed GD: Motivation



- Consider $f(x, y) = 0.5(x^2 - y^2)$.
  If we initialize at $(0, 0)$ GD will not move.
- However, the saddle point is instable and if we
  inject random noise in the gradient, GD will eventually escape it

$$x_{t+1} = -\eta \sum_{\tau=1}^{t} (1-\eta)^{t-\tau} e_\tau(x), \qquad y_{t+1} = -\eta \sum_{\tau=1}^{t} (1+\eta)^{t-\tau} e_\tau(y) \tag{14}$$

  where $e_\tau(x)$ and $e_\tau(y)$ are the random noise
  added to the x and y component of the gradient, respectively.
- Remark: SGD has baked in noise, but explicit injection is needed
  in general.
- Remark: With random initialization, the chance of starting at a
  bad initialization (saddles and local maxima) is very low

Perturbed GD



- In each iteration run

$$w_{t+1} = w_t - \eta \nabla f(w_t) + \eta e_t, \qquad e_t \sim S(0,1) \quad \text{or} \quad e_t \sim B(0,1) \quad \text{or} \quad e_t \sim \mathcal{N}(0, I) \tag{15}$$

- Essentially, we use random exploration to find a direction of decrease, as opposed to using second order information (Hessian)
- Related to Langevin Dynamics

Theorem

Assuming L-first order and ρ-second order smoothness, perturbed GD can find an $(\epsilon, \sqrt{\epsilon})$ solution with high probability when $T = \Omega(d^2 \epsilon^{-1})$.

- Recall CRNM needs $T = \Omega(\epsilon^{-0.75})$ deterministically and w/o requiring L-first order smoothness
- The issue is adding noise in every single iteration, even if we are not near a saddle (seems unnecessary)

Better Perturbed GD



- Only add noise when needed (near a first-order solution that has a low gradient norm)
- for any t if $\|\nabla f(w_t)\| \leq \kappa$
  - $w_t \leftarrow w_t + e$ with $e_t \sim B(0, r)$ with $r = \mathcal{O}(\epsilon)$
- Do GD for $\tilde{T}$ iterations

Theorem

Assuming L-first order and ρ-second order smoothness, better perturbed GD can find an $(\epsilon, \sqrt{\epsilon})$ solution with high probability when $T = \tilde{\Omega}(\epsilon^{-1})$.

- Finding 2nd-order solutions is almost as easy as finding first-order solutions (when ignoring log terms that may depend on dimensions)